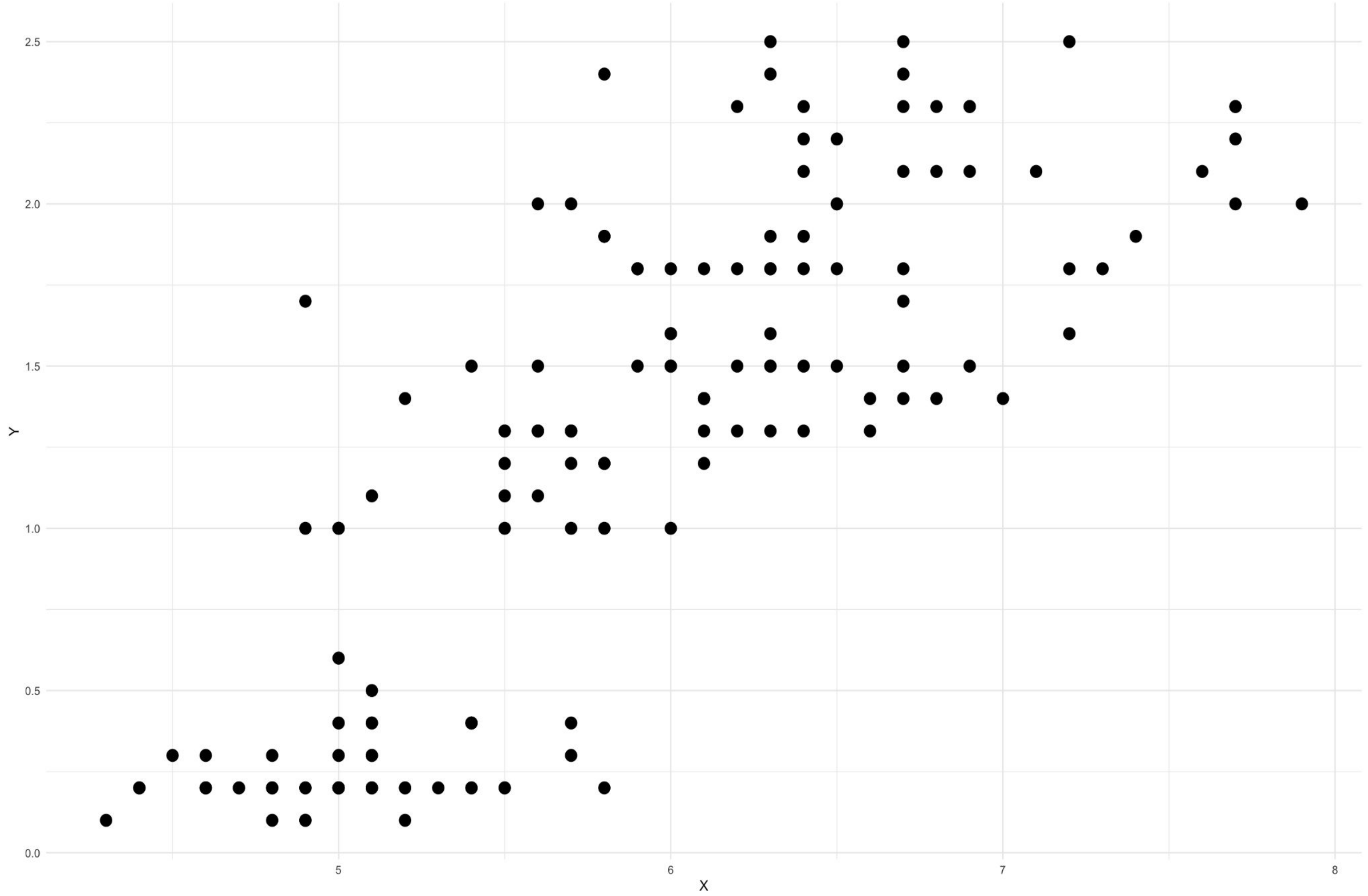
Clusterization
K-means

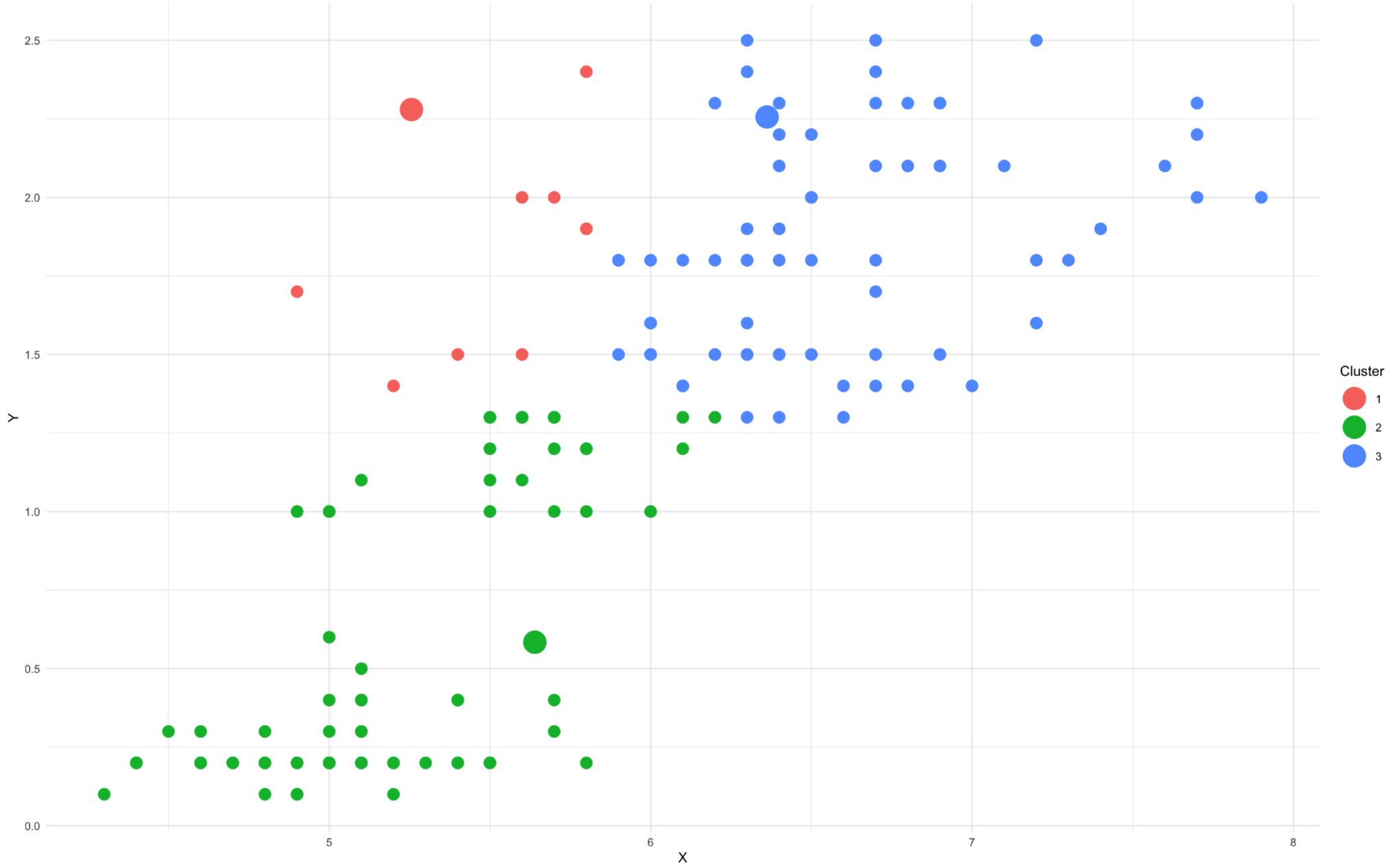
Clusters in iris dataset

- Let’s see if there are clusters in sepal length against petal width from iris dataset
- Let k = 3 (there are actually 3 clusters)

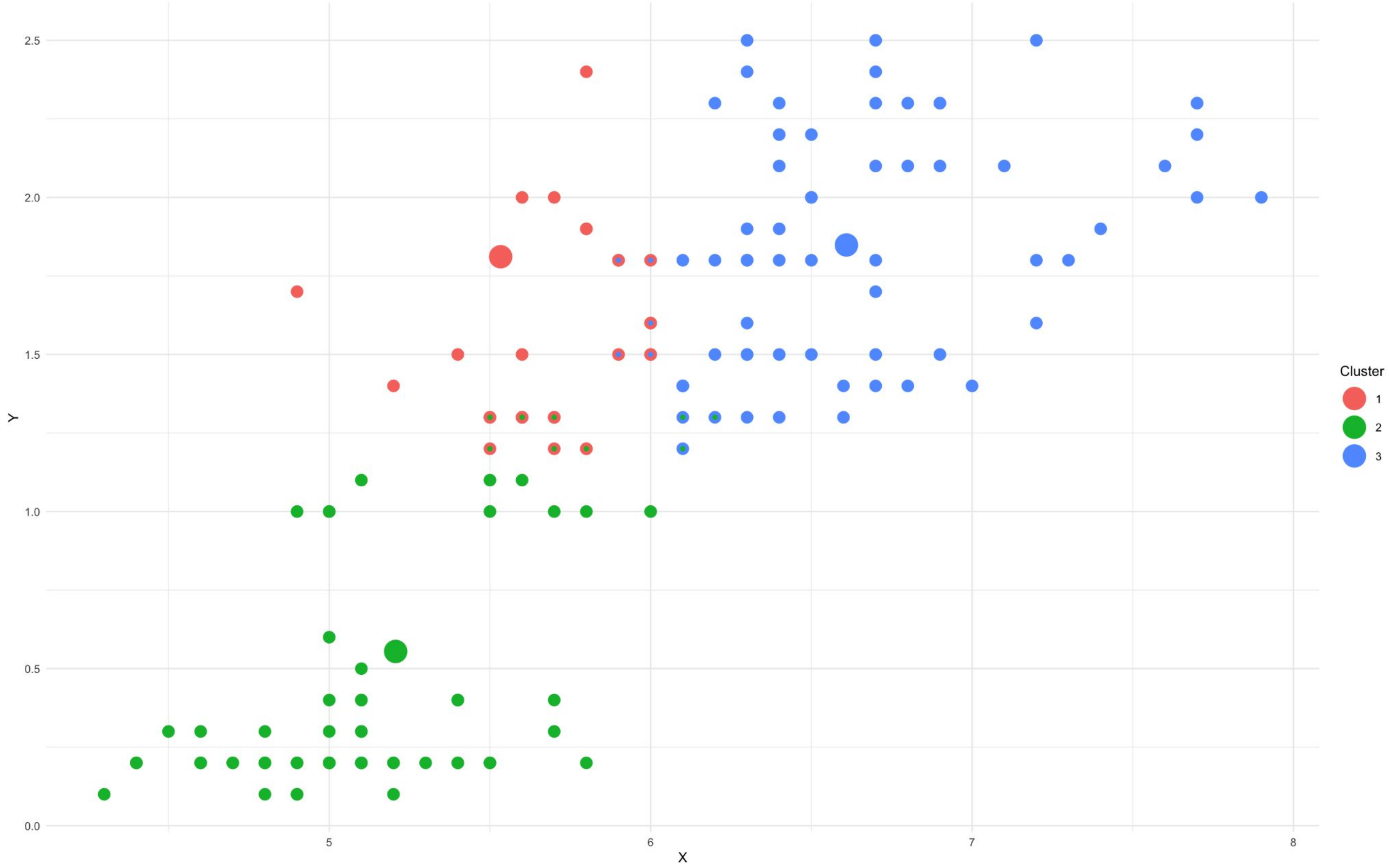
```
set.seed(1)
k_means_visual(k = 3,
               data = iris[,c("Sepal.Length","Petal.Width")],
               print_plot = TRUE)
```



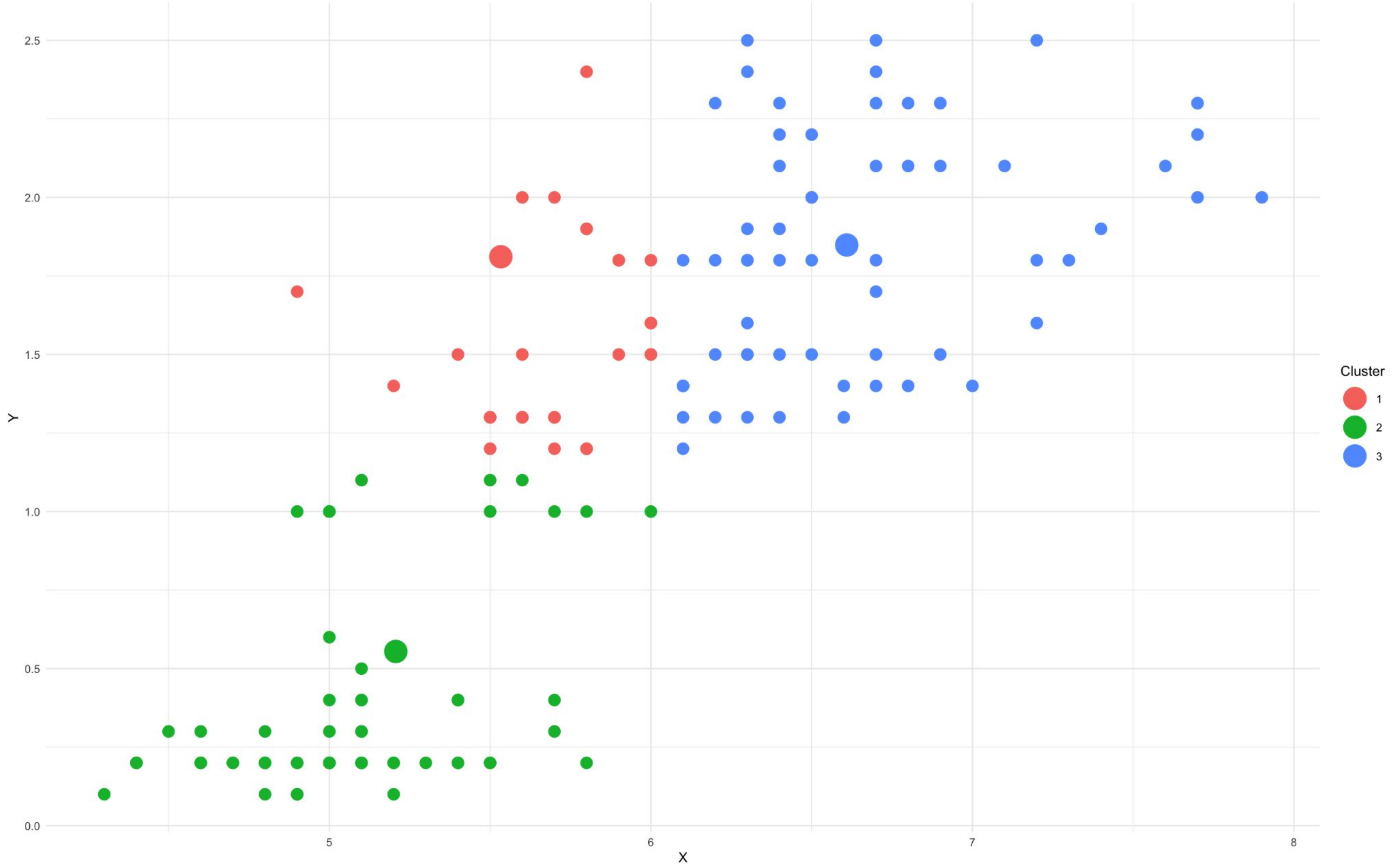
2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X



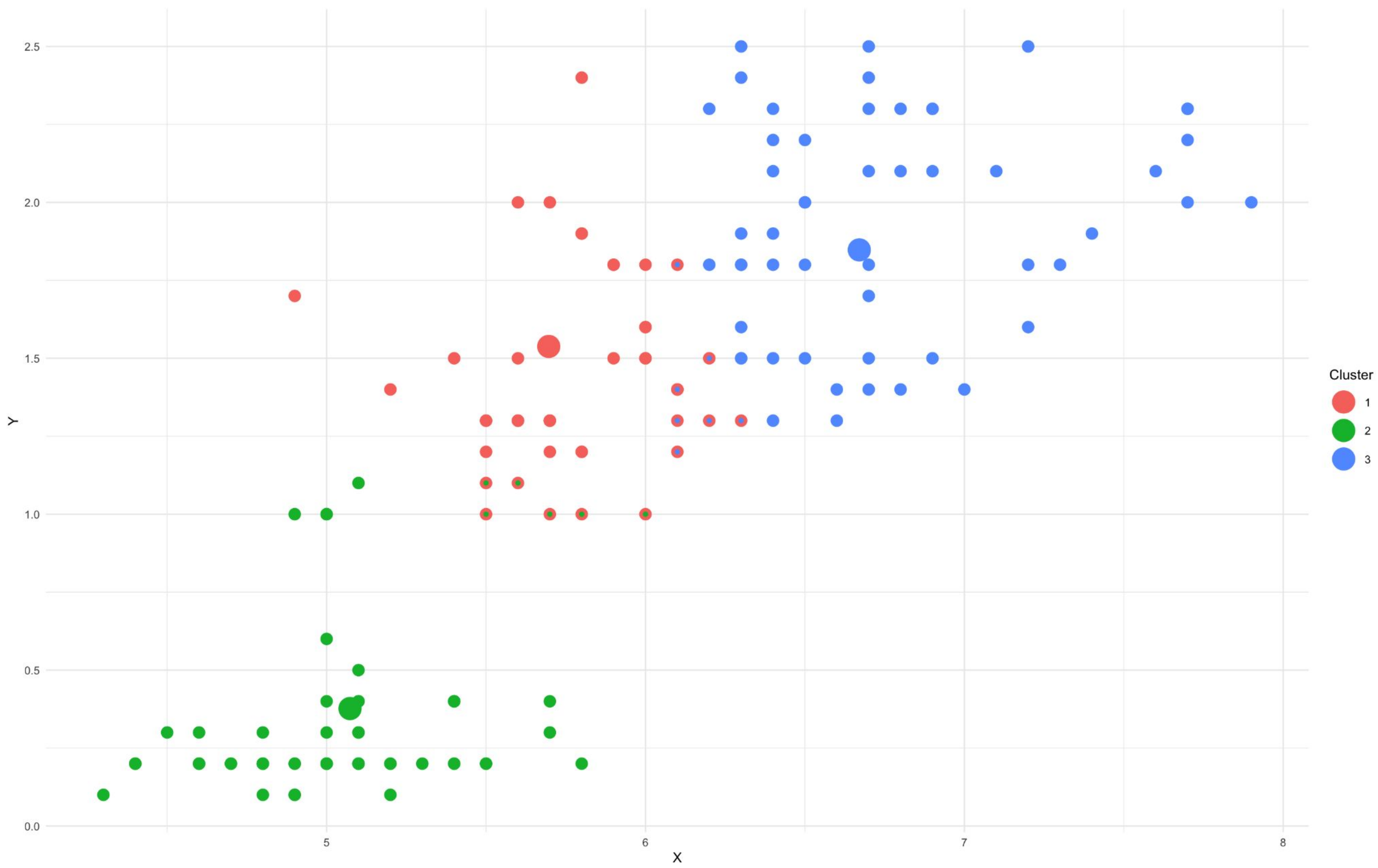
2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

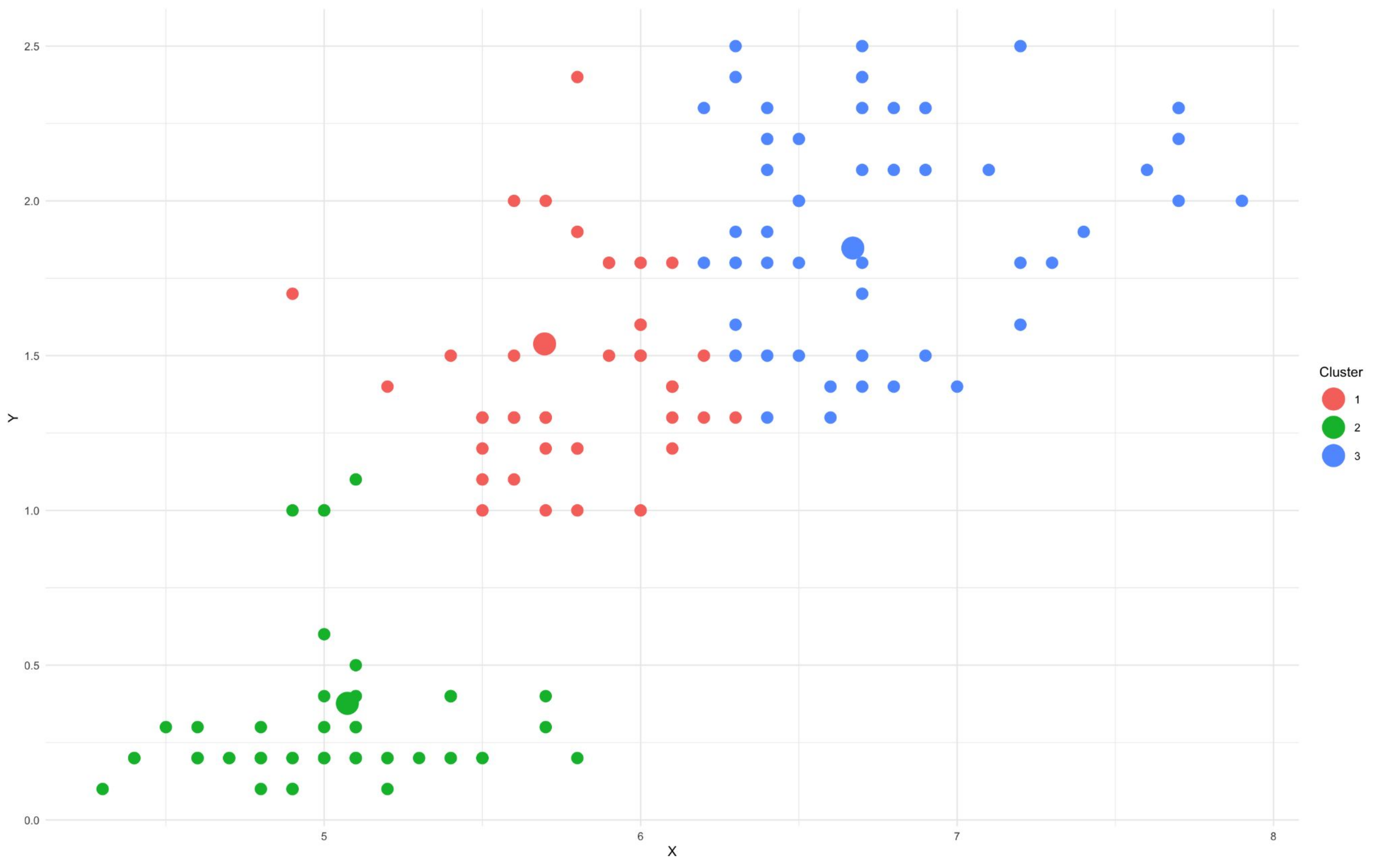


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

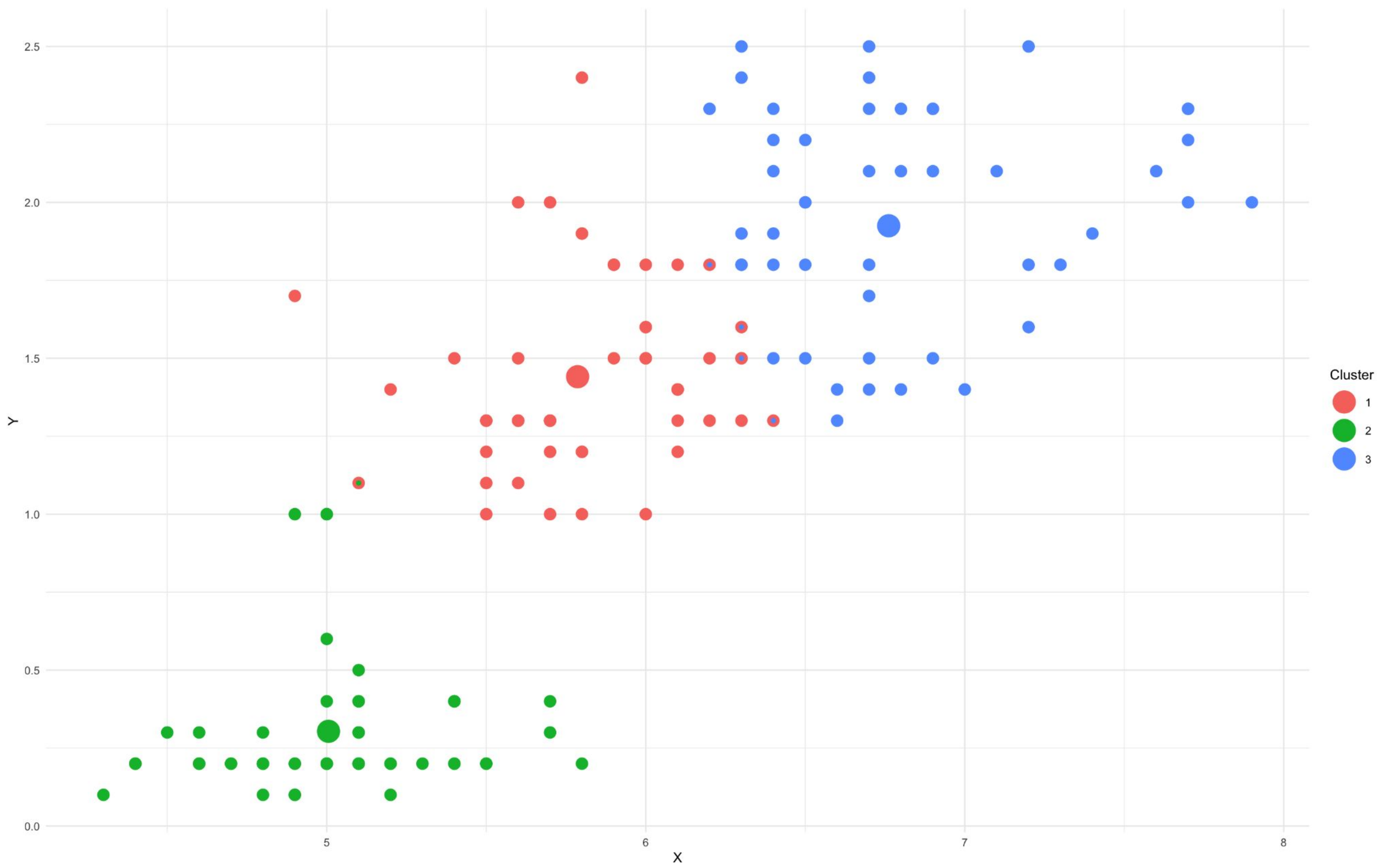


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

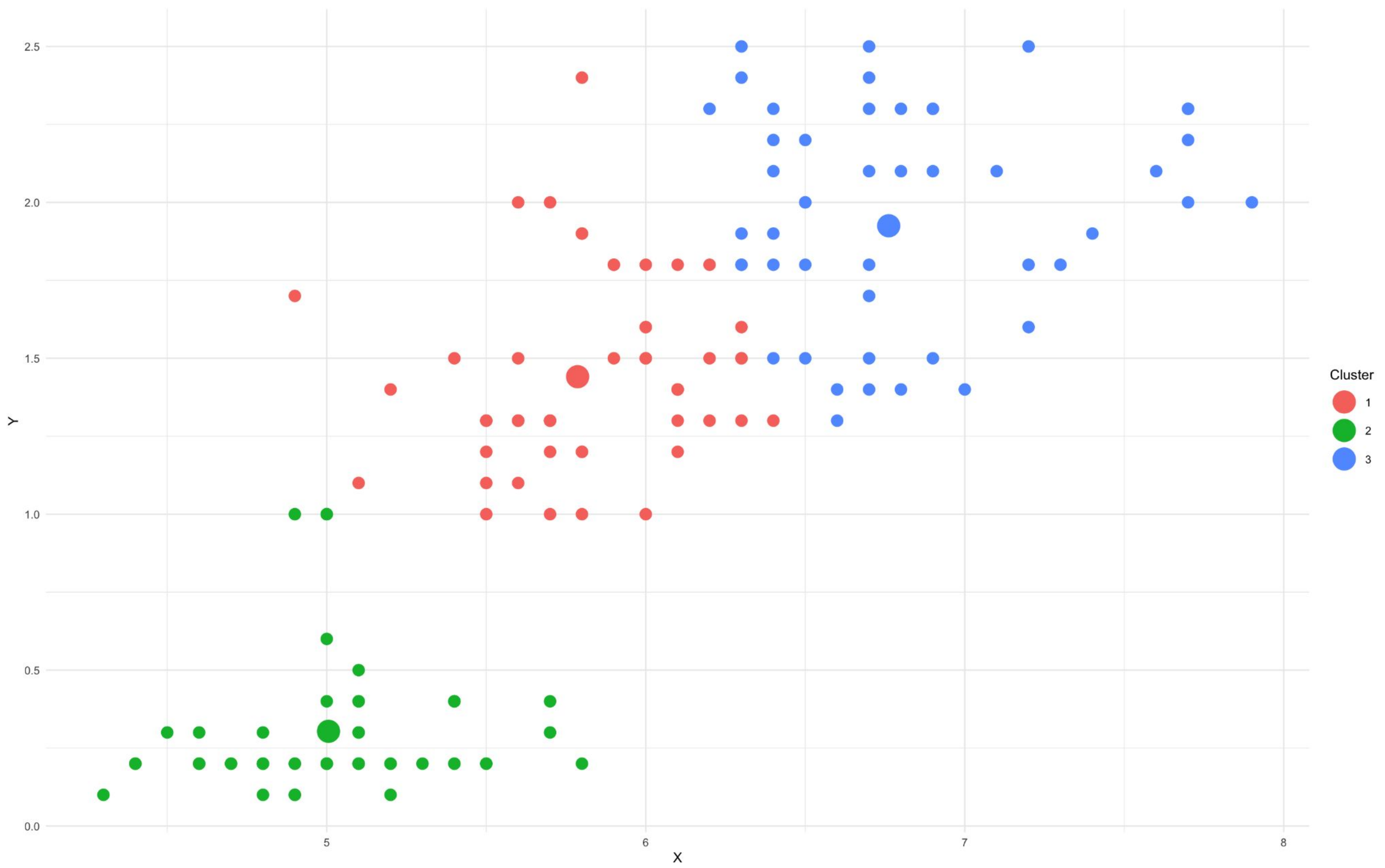

2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

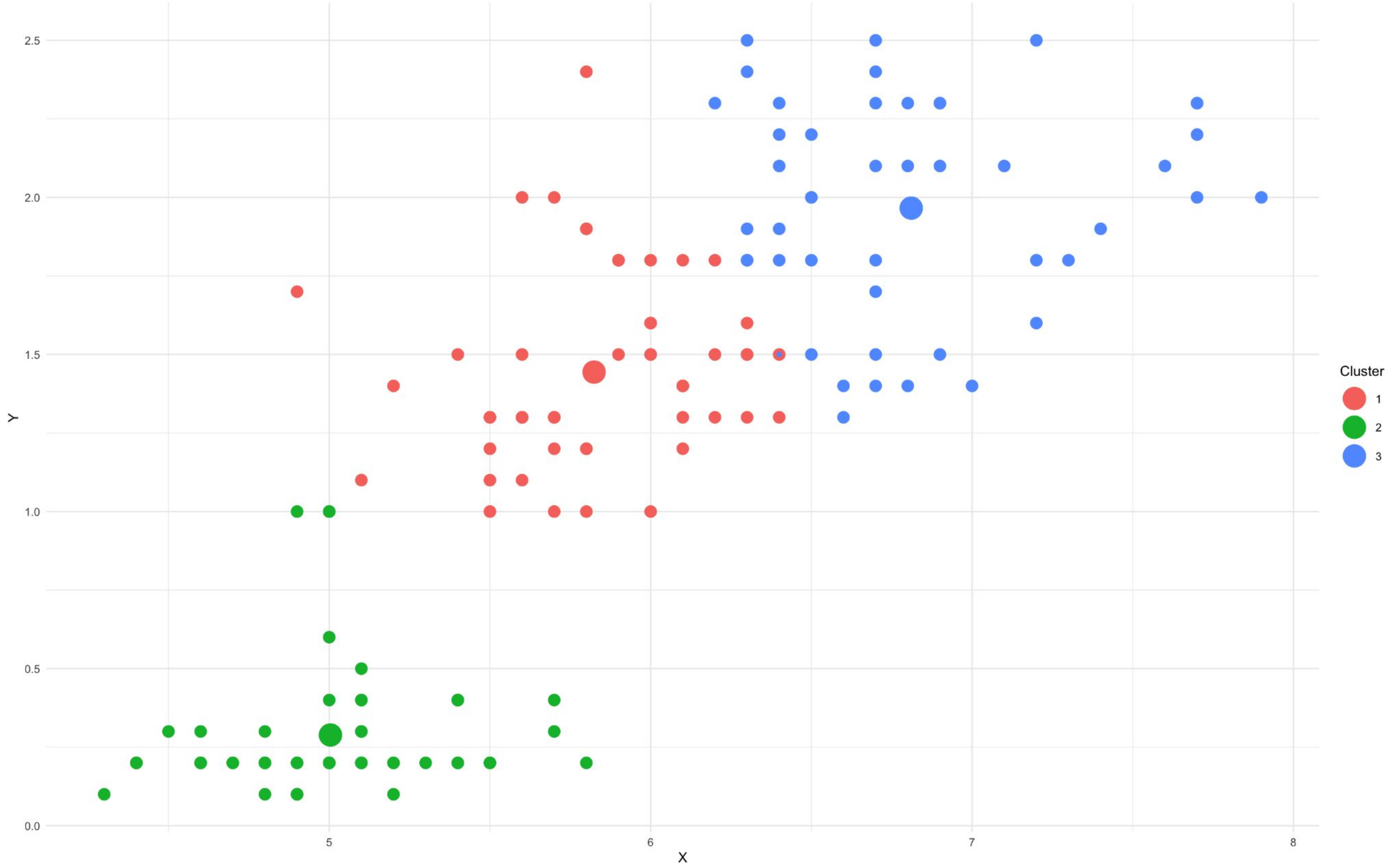


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

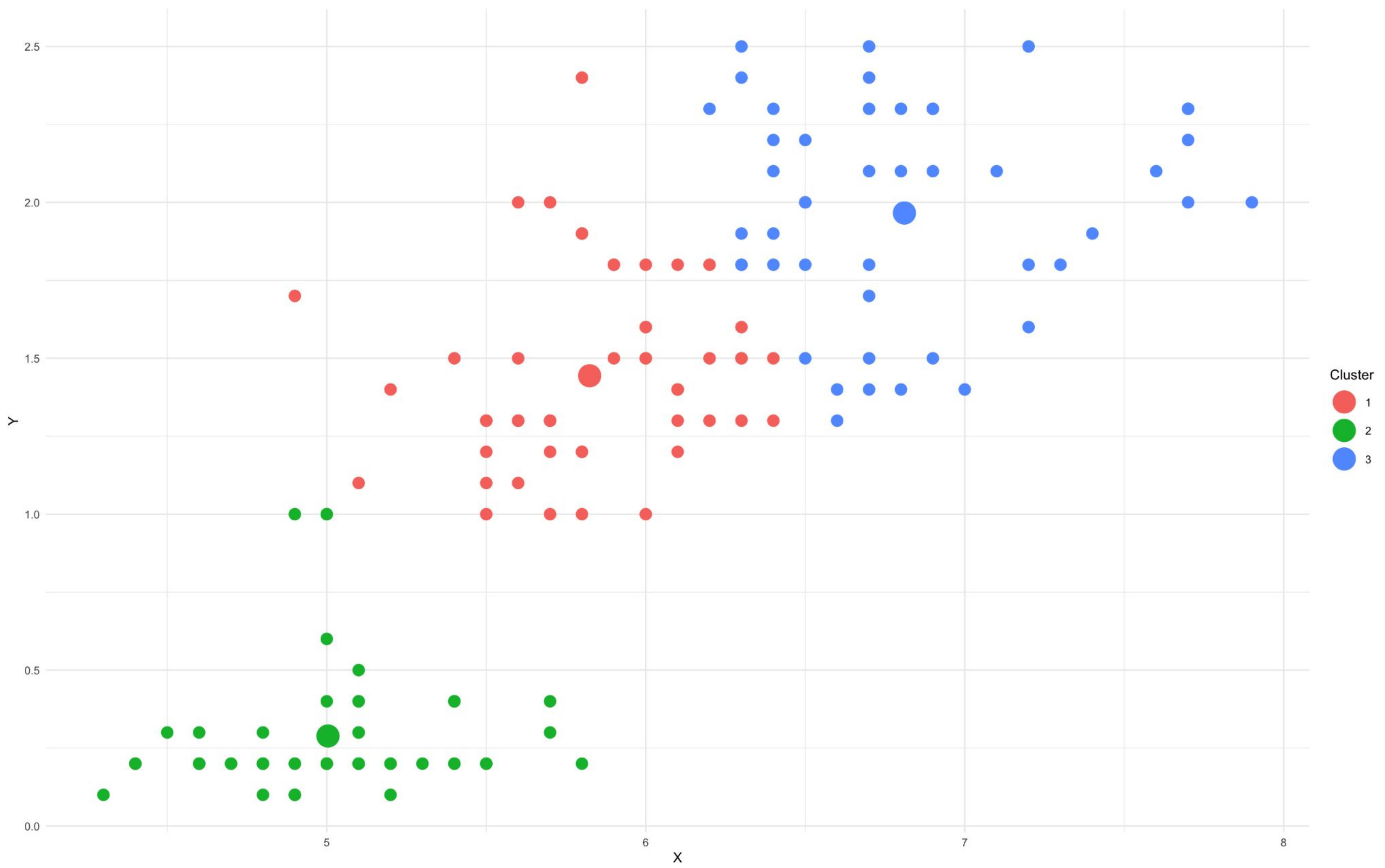


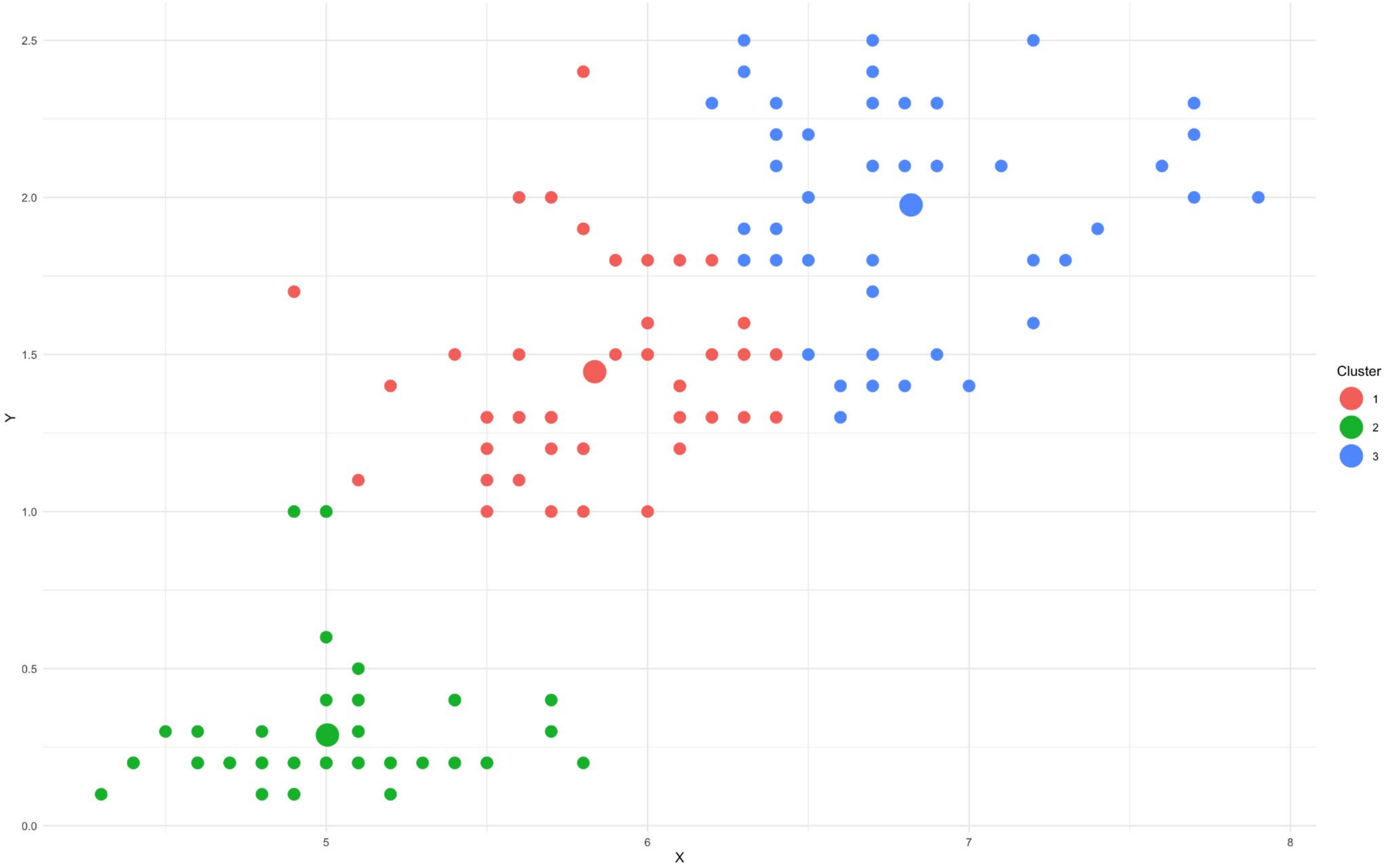
2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

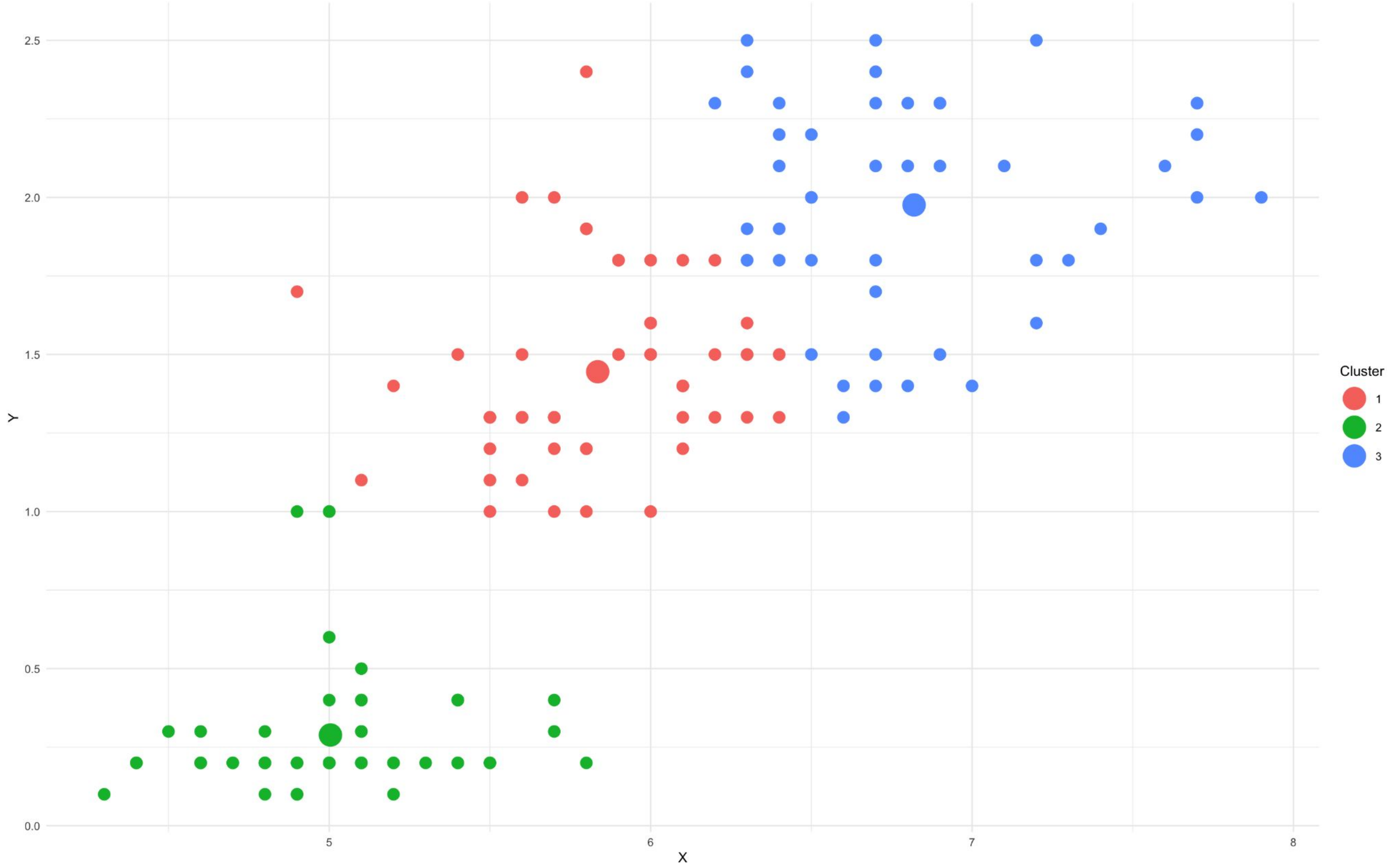


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3


2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3



2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3



2.5
2.0
1.5
1.0
0.5
0.0
Y
5
6
7
8
X
Cluster
1
2
3

Output

```
$clusters
  [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 [37] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 1 3 1 3 1 1 2 3 1 2 1 1 1 1 3 1 1 1 1 1 1
 [73] 1 1 1 3 3 3 1 1 1 1 1 1 1 1 3 1 1 1 1 1 1 2 1 1 1 1 1 1 3 1 3 3 3 3 1 3
[109] 3 3 3 3 3 1 1 3 3 3 3 1 3 1 3 3 3 3 1 1 3 3 3 3 3 1 1 3 3 3 1 3 3 3 1 3
[145] 3 3 3 3 3 1

$cluster_locations
            x         y
[1,] 5.835294 1.4450980
[2,] 5.003774 0.2886792
[3,] 6.819565 1.9760870
```

Best k

- Let’s now find the best k value using total within-cluster sum of squares
- Let’s test k’s between 1 and 10

```
set.seed(1)
optimal_k(max_k = 10,
          data = iris[,c("Sepal.Length","Petal.Width")])
```